Exploratory Data Analysis and Visualization

Animal Classification

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Dataset Dimensions

The dataset dimensions of (14377, 3) imply there are three columns, representing:

1. **Image Path**: The file path location of each image.
2. **Label**: The associated class label for the image.
3. **Type**: This likely refers to a secondary attribute or category for the image, such as whether it belongs to a certain category, dataset split (train/test), or another classification type.

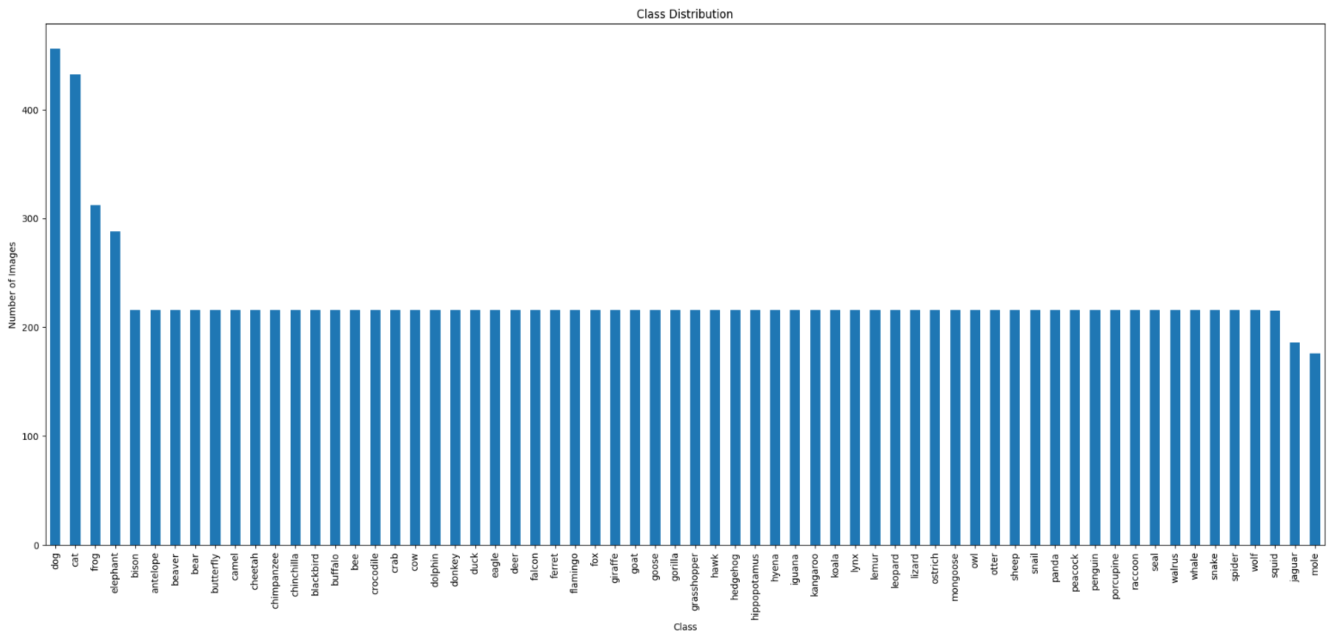
These columns provide the essential structure for training a machine learning model, linking each image with its label and type.

Univariate Analysis

Class Distribution

This technique provides a clear view of how the instances (images, in this case) are distributed across different classes in a dataset. It is crucial for understanding the balance or imbalance in the dataset, which can affect the performance of machine learning models. Here's why it's useful:

- **Identifying Class Imbalance:** If one class dominates the dataset, it may lead to a biased model that performs poorly on underrepresented classes.
- **Model Performance Prediction:** A balanced dataset usually leads to better generalization and performance, while an imbalanced dataset might require techniques like oversampling, under sampling, or class-weight adjustment.



The image shows two bar charts: one for the training set and the other for the test set class distribution.

- **Training Set**: Classes have about 160-170 samples, with a noticeable peak of around 350 samples for the class "dog".
- **Test Set**: Classes are more evenly distributed, with each class having about 40-45 samples, and a peak around 90 samples for " dog ".

Conclusion: The dataset is balanced in both training and test sets. The slight peak for "dog" in both sets does not significantly affect the fairness of model evaluation, suggesting a well-distributed dataset for model training and testing.

Insights and Findings

1. Imbalanced Classes:

The "dog" and "cat" classes have a notably higher number of images compared to the other classes. The “frog” and “elephant” have slightly lower number of images as compared to “cat” and “dog” but higher number of images as comapred to other classes.This indicates an imbalance in the dataset, which could potentially affect the performance of a machine learning model. Specifically, the model may become biased towards these overrepresented classes, as it would see them more frequently during training.

2. Relatively Uniform Distribution:

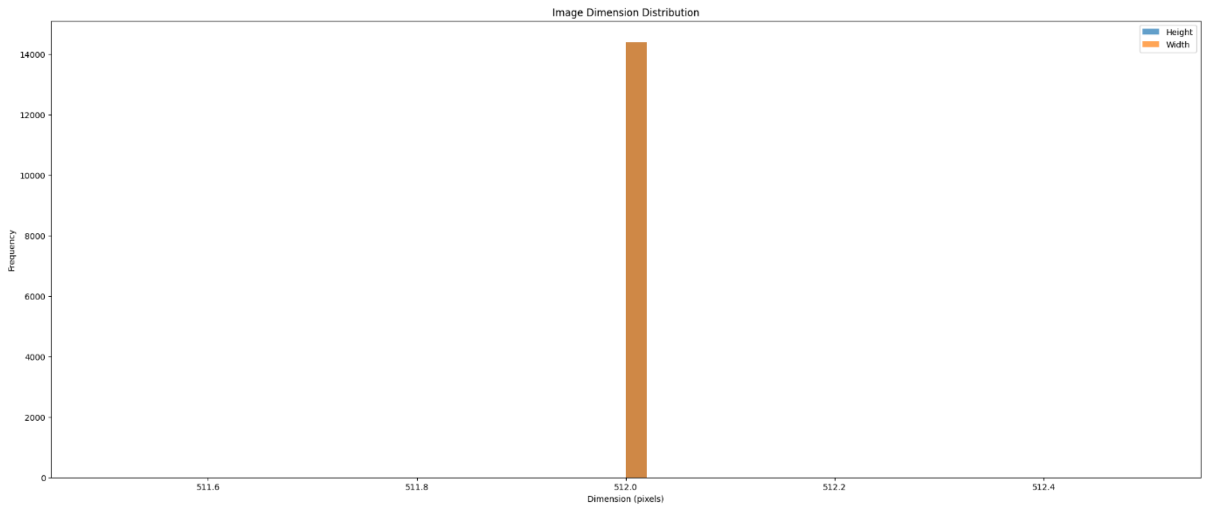
Most of the other classes exhibit a fairly uniform distribution of images. This is generally beneficial for model training, as a more balanced dataset allows the model to learn equally from all classes. However, there are still some variations in the number of samples for these classes.

3. Minor Classes with Slightly Fewer Images:

Certain classes, such as "jaguar," and "mole" appear to have slightly fewer images, making them underrepresented in the dataset. This underrepresentation could lead to the model struggling to learn characteristics specific to these classes.

Image Dimension Variability

This method is focused on analyzing the dimensions of images within a dataset. The primary objective is to understand the distribution of image heights and widths by visualizing their frequencies. The analysis involves calculating the height and width of each image, followed by plotting histograms to illustrate the distribution.



Insights and Findings

1. Uniformity in Image Dimensions:

The histogram reveals that most images in the dataset have consistent dimensions, centered around 512 pixels for both height and width. This indicates a uniformity that simplifies preprocessing requirements for model training.

2. Lack of Variability:

The absence of significant variations in image sizes, as shown by the histogram, suggests that the dataset is well-suited for image classification tasks. Minimal resizing is needed, reducing the risk of distortion during preprocessing.

3. Dataset Preparedness:

The consistent dimensions, with heights and widths being approximately equal, imply that the dataset might already be preprocessed to square dimensions. This ensures easier handling and preparation of images for input into deep learning models.

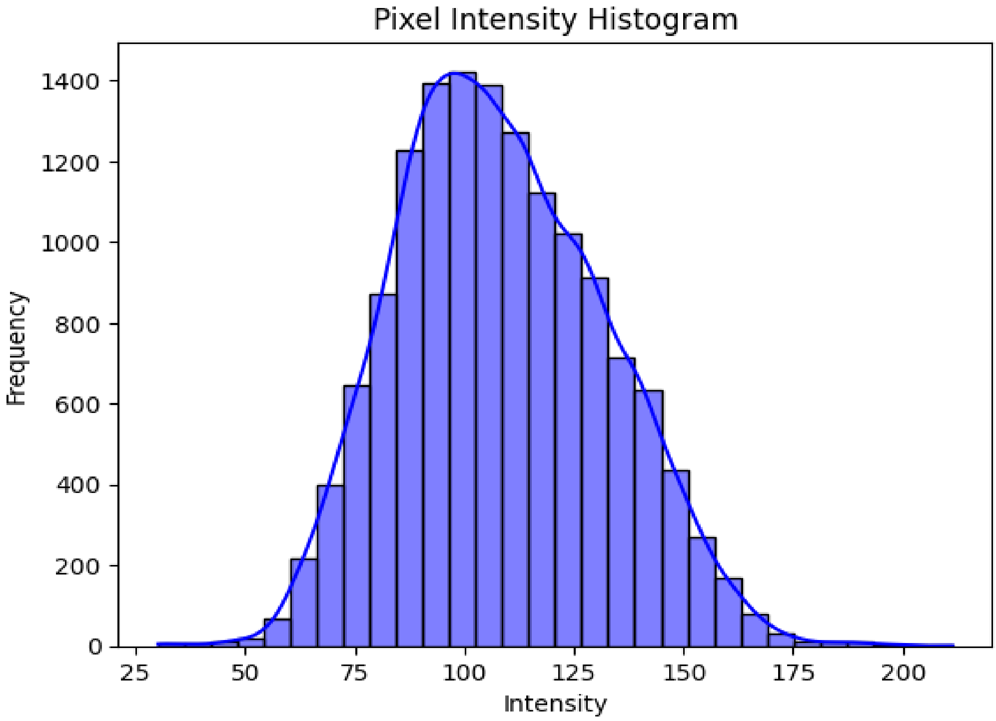
Relevance to Animal Image Classification:

- **Ease of Training**: The uniform image dimensions aid in standardizing the inputs for neural networks, which simplifies the training process and reduces preprocessing complexity.

- **No Resize Needed**: The consistency in dimensions means that resizing operations might not be required unless the model input size is different. This enhances the efficiency of model preparation.

Pixel Intensity Distribution

This method provides a detailed analysis of the average pixel intensity across all images in the dataset. The average pixel intensity is calculated as the mean of all pixel values (across RGB channels) in an image. This technique is useful for evaluating the overall brightness levels and detecting potential inconsistencies in the dataset.



Insights and Findings

1. **Gaussian-like Distribution**
   The pixel intensity distribution shows a bell-shaped curve, with most images having average intensities between 75 and 125. This indicates that the dataset is well-balanced in terms of brightness, without a significant skew toward either darker or brighter images.

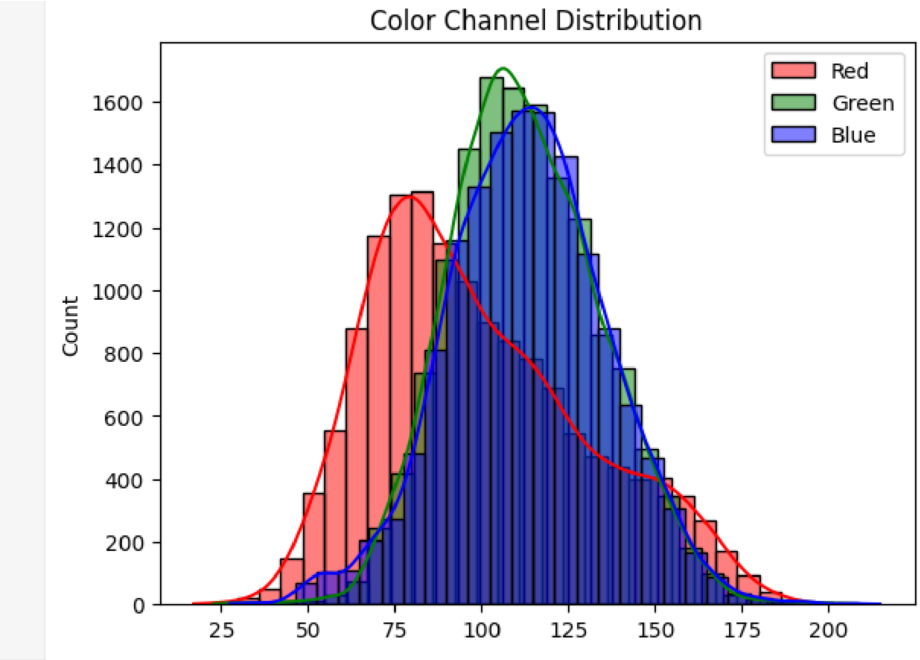
2. **Presence of Outliers**

- A few images with low average intensities (around 25) suggest underexposure or darkness, which could lead to poorer feature learning for these images.
- Similarly, there are a few images with high average intensities (over 200), indicating overexposure or excessive brightness. These outliers may require correction or exclusion during preprocessing.

3. **Balanced Brightness Levels**
   The overall distribution suggests that the dataset has consistent brightness levels for the majority of images, which is beneficial for machine learning models. A balanced dataset ensures that the model does not disproportionately favor certain brightness ranges during training.

4. **Preprocessing Recommendations**
   - **Normalization**: Pixel values should be scaled to a uniform range (e.g., 0–1) to standardize brightness levels across all images.
   - **Outlier Handling**: Extremely dark or bright images may need brightness adjustments, such as gamma correction or histogram equalization, to bring them in line with the rest of the dataset.

Color Channel Distribution Analysis



Insights and Findings

1. **Distinct Peaks in Channels**

   - **Red Channel**: Shows a peak around 75–80, indicating dominance of medium red hues in the dataset. Fewer images exhibit high-intensity reds.
   - **Green Channel**: Peak around 100 with a gradual drop-off. Green hues are relatively more balanced and prevalent.
   - **Blue Channel**: Peak around 125 with a slow decline, suggesting a prevalence of images with brighter blue tones.

2. **Distribution Balance**

   The dataset appears balanced across color channels, without extreme skew toward any specific color, ensuring uniform representation.

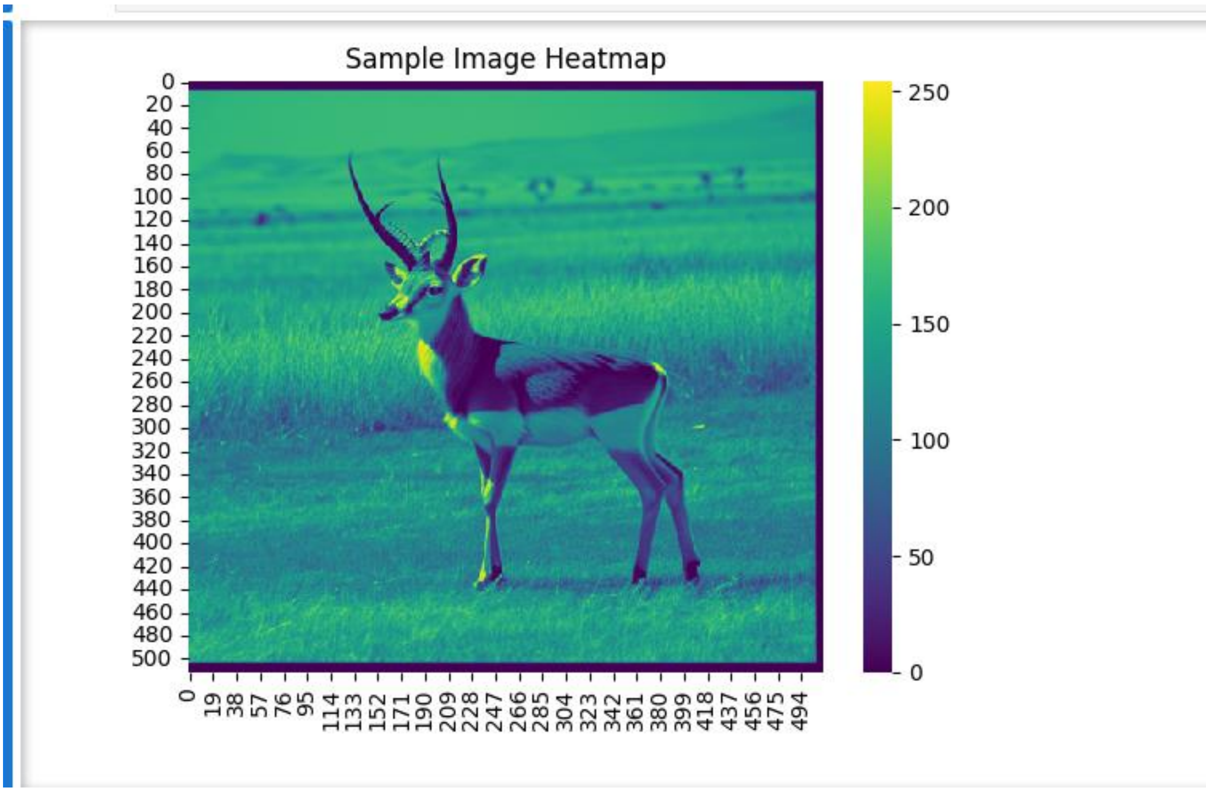
3. **Outlier Detection**

   - Few images with extreme values in one or more channels indicate potential anomalies (e.g., overly red, green, or blue-dominant images).
   - Such outliers may require inspection for preprocessing adjustments.

4. **Preprocessing Recommendations**
    - **Channel Normalization:** Standardize all channels to a consistent scale (e.g., 0–1).
    - **Outlier Handling**: Perform histogram equalization or color adjustments to ensure uniform color intensity distribution.

HeatMap Analysis

Insights and Findings



1. **Observations**

   - **Intensity Distribution:** The heatmap visualizes pixel intensity values, with brighter regions indicating higher intensity and darker regions indicating lower intensity.
   - **Object and Background Segmentation:**
     1. The primary subject (likely an animal, such as a deer or antelope) appears as a brighter region, highlighting higher intensity values.
     2. The background (e.g., grass or trees) is represented by darker areas, showing lower intensity values.

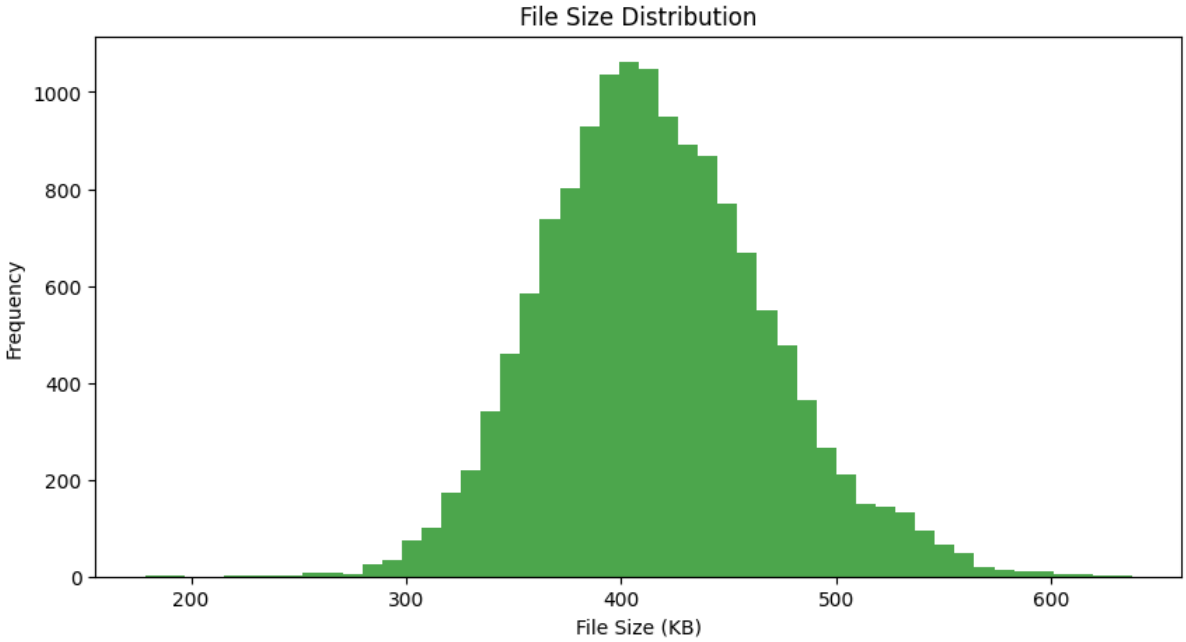
2. **Possible Applications**

   - **Image Segmentation**: Identifies distinct regions (object vs. background).
   - **Object Detection**: Pinpoints the object's location.
   - **Image Recognition**: Analyzes patterns for classification tasks.
   - **Machine Learning Input**: Provides data for training models to enhance detection or recognition accuracy.

File Size Distribution

This method evaluates the distribution of file sizes (in kilobytes) for all images in the dataset. File size is a critical factor in data quality and storage considerations, as it can indicate potential issues like corruption or inconsistency in image resolution or compression. Here's why this analysis is important:

- **Corrupted Files**: Extremely small file sizes may indicate corrupted or missing images.
- **Abnormally Large Files**: Larger file sizes could point to high-resolution images or poor compression, which might impact processing speed and storage requirements.
- **Dataset Uniformity**: A consistent file size distribution often indicates uniformity in image dimensions and compression levels, which is beneficial for downstream machine learning tasks.



Insights and Findings

1. **Gaussian-like Distribution**
   The file size distribution follows a bell-shaped curve, with most images having file sizes between 300 KB and 500 KB. This indicates that the majority of the dataset is consistent in terms of resolution and compression quality.
2. **Presence of Outliers**

- **Smaller Files**: A few images have sizes below 200 KB, which may indicate corruption, missing data, or extremely low-resolution images. These files should be reviewed and potentially excluded or replaced.
- **Larger Files**: Some images exceed 600 KB, suggesting high-resolution or less-compressed files. These might introduce computational inefficiencies during training and may need resizing or re-encoding.

3. **Balanced File Sizes**
   The majority of files are clustered within a relatively narrow range, demonstrating that the dataset is generally consistent in terms of size. This uniformity supports efficient data loading and preprocessing.
4. **Preprocessing Recommendations**
   - **Outlier Handling**: Identify and review files with abnormally small or large sizes to ensure data quality. Corrupted files should be removed, and oversized files should be resized to match the majority of the dataset.
   - **Standardization**: Resizing all images to a consistent resolution (e.g., 256x256 or 512x512) can help ensure uniform file sizes and improve processing efficiency.

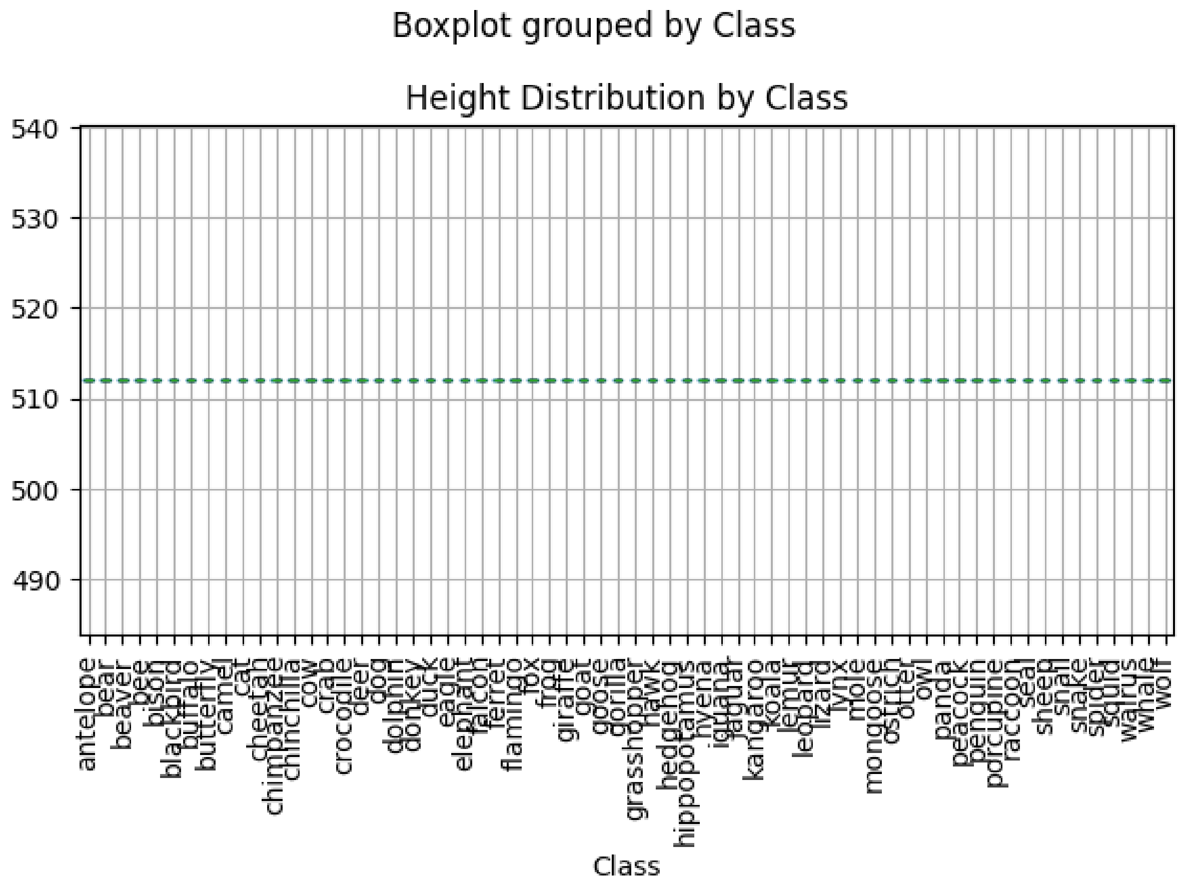
Bivariate Analysis

Height Distribution by Class

This plot offers a view of how the **height** variable is distributed for each class, providing insights into the variability and central tendency of height within each animal class. The visualization aids in identifying classes with a greater range of height values and those with more consistent height measurements. Here's why this information is valuable:

- **Identifying Variability in Heights**: By analyzing the range, median, and presence of outliers for each class, this plot helps to understand which classes have more variability in height. This can provide clues for modeling and help avoid assumptions of uniform distribution across classes.

- **Outlier Detection**: Classes with extreme height values can indicate potential outliers, which might be interesting for further examination or may need handling in a machine learning context.



Insights and Findings

1. **Height Consistency Across Classes**:
   Many classes show a consistent height distribution, with relatively compact interquartile ranges (IQR) and few outliers. This suggests a uniform height distribution across these classes.

2. **Classes with High Height Variability**:
   Some classes display a wider range in height, indicating high variability. This could mean a more diverse representation of heights within these classes, which

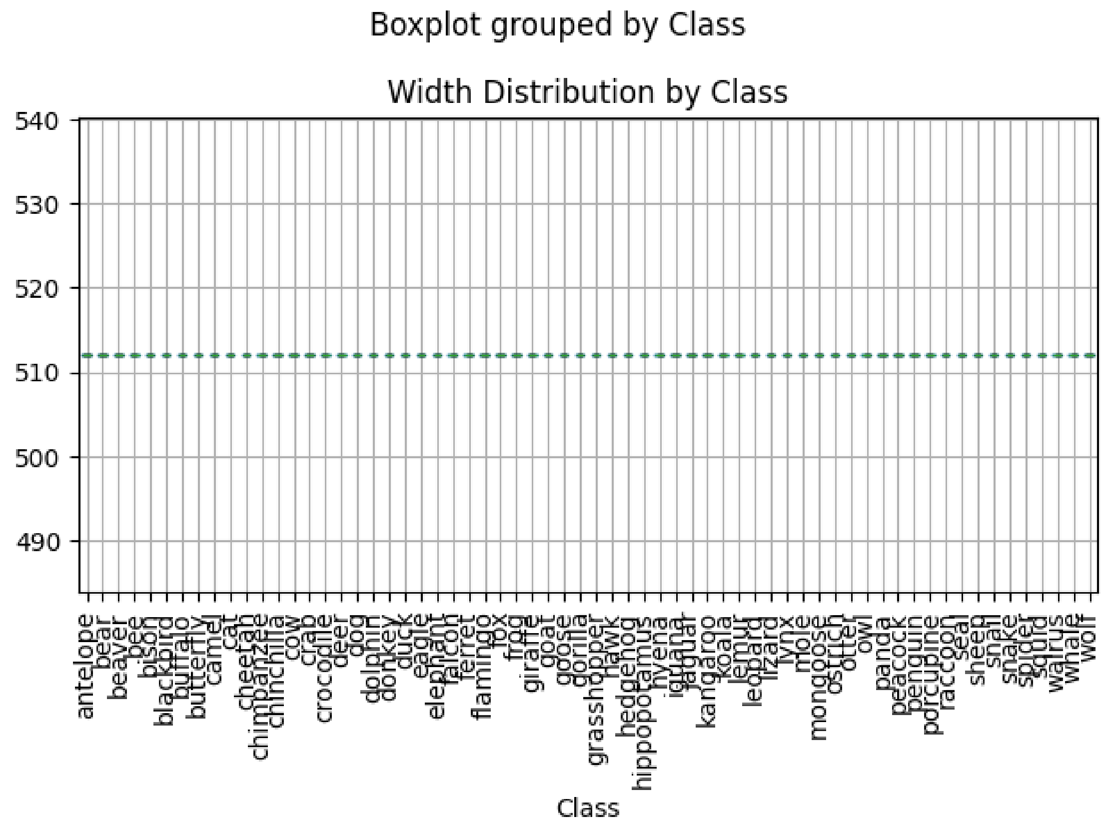
may affect the model's ability to generalize across all height values within a single class.

3. **Outliers and Extreme Values**:
   The presence of outliers in certain classes suggests that some instances deviate significantly from the median height. This could impact a model trained on these data, as these extreme values might need special consideration to avoid skewing the model's performance.

Width Distribution by Class

This analysis provides insights into the distribution of the **Width** variable across different classes in the dataset. Understanding the width distribution helps identify variations within and between classes, which is important for model training and performance. Here's why this analysis is useful:

- **Identifying Class Variations**: Differences in width distributions across classes can highlight class-specific features or biases in the dataset.
- **Improving Model Generalization**: Uniform distributions and balanced features allow for better model learning, while wide variability or outliers may require handling through preprocessing.



Insights and Findings

1. **Consistent Width Distribution in Certain Classes**:

   Some classes exhibit a narrow interquartile range indicating consistent width values. This uniformity can be beneficial for training but might limit the model's ability to generalize if the dataset is biased.

2. **Significant Variability in Other Classes**:

   Some classes show a wider distribution of widths, with larger interquartile ranges. This suggests greater variability in these classes, which could reflect the natural diversity in these categories.

3. **Presence of Outliers**:

   Several classes display outliers in the boxplot. These points represent unusually small or large widths, which might impact the training process if not addressed.

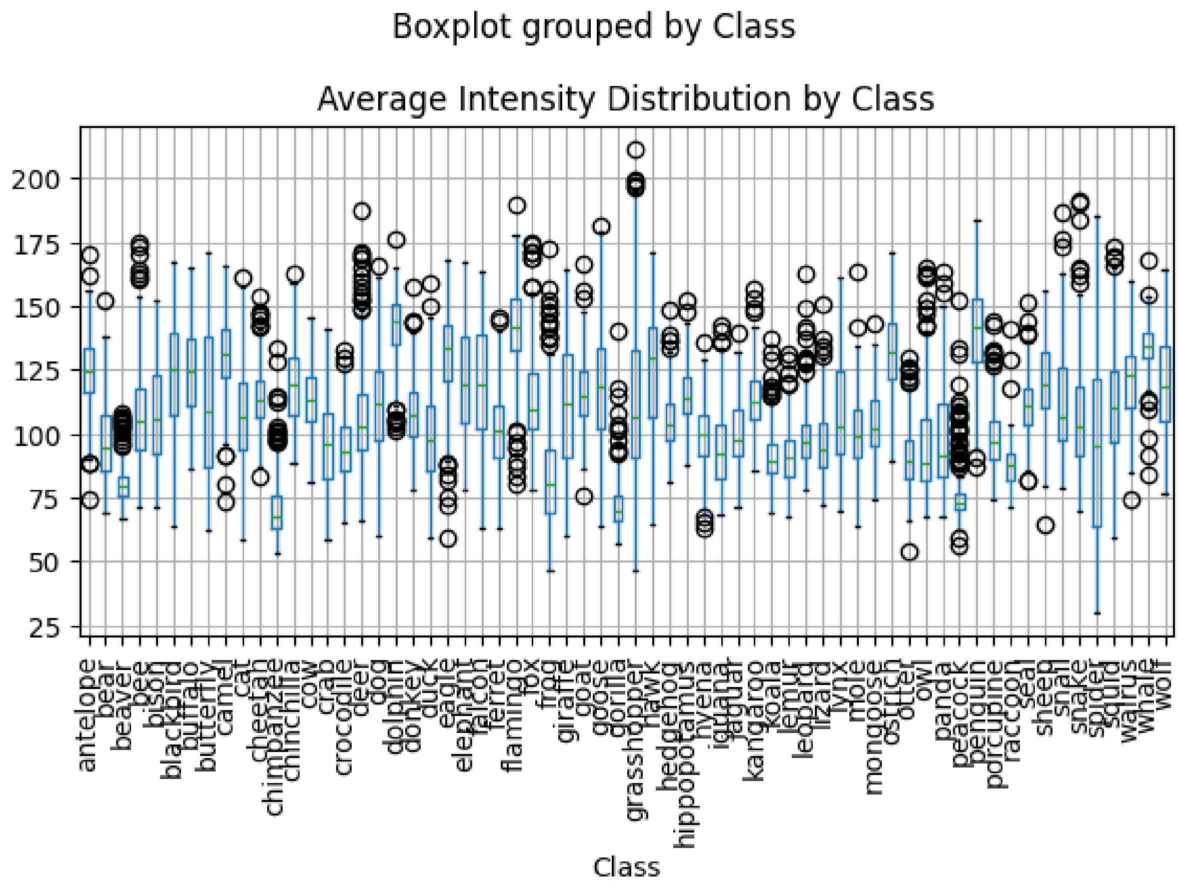
4. **Potential Imbalance Across Classes**:

   While the width distribution appears relatively uniform for many classes, certain classes with larger ranges or multiple outliers may dominate the dataset. This could lead to biased learning, favoring classes with wider variability.

Average Intensity Distribution by Class

This analysis examines the distribution of **Average Intensity** for each class in the dataset. The intensity values provide important insights into the brightness or pixel intensity variations across different classes, which can be crucial for distinguishing between them. Here's why this analysis is significant:

- **Understanding Feature Variability**: Intensity distributions can highlight whether certain classes have distinct or overlapping brightness characteristics.
- **Improving Classification Models**: Identifying unique intensity patterns can aid in building more accurate models by leveraging these features effectively.

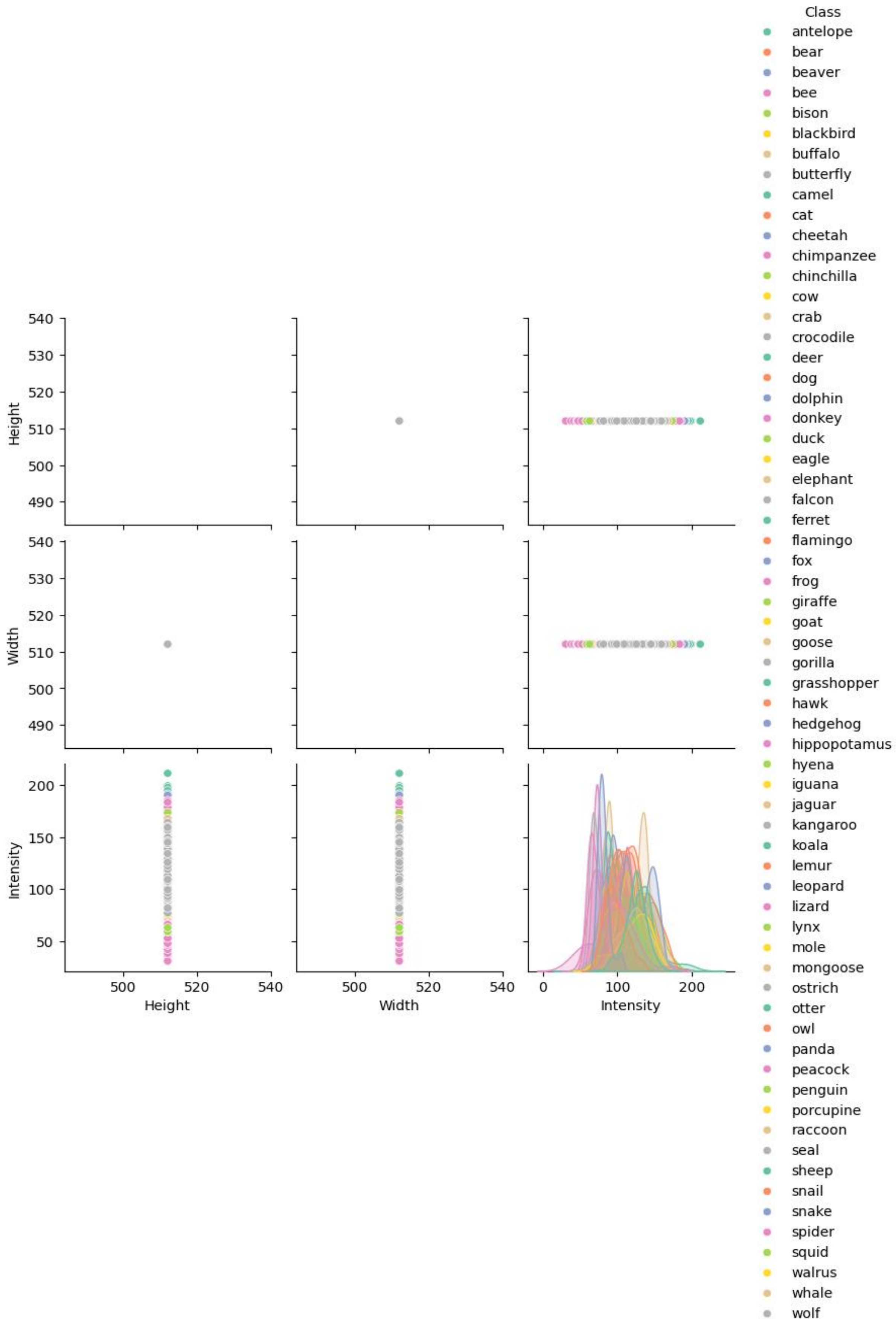


Insights and Findings

1. **Wide Intensity Ranges in Certain Classes**:
   Classes such as "butterfly" and "falcon" display a broad range of average intensity values, reflecting significant variability in brightness. This suggests high diversity within these classes, which may challenge the model's ability to generalize.
2. **Consistent Intensity in Some Classes**:
   Classes like "cow" and "bear" show narrower interquartile ranges, indicating consistent intensity values. Such uniformity can provide a stable feature for classification but might reduce model adaptability for these classes.
3. **Presence of Outliers**:
   Several classes have outliers, with intensity values significantly higher or lower than the main distribution. For instance, "deer" and "frog" show extreme values. These may represent noise, lighting conditions, or unique class-specific characteristics that require careful handling.
4. **Uniformity Across Most Classes**:
   Despite some variability, most classes demonstrate a relatively balanced intensity distribution without extreme skews. This balance is beneficial for ensuring that no single class dominates the feature space.

Pairplot Analysis: Relationships Between Height, Width, and Intensity by Class

This analysis uses a pairplot to explore the relationships between the features **Height**, **Width**, and **Intensity** across different animal classes in the dataset. By visualizing these pairwise relationships, we aim to understand feature distributions and class separability.



Insights and Findings

1. **Distinct Patterns in Intensity:**

    - The Intensity feature demonstrates considerable variability among the classes, as seen in the density plots. Some classes show distinct intensity distributions, while others overlap. This suggests that intensity might be a useful feature for distinguishing between certain classes.

2. **Overlap in Classes:**

    - Despite variability in Intensity, there seems to be some overlap in distributions between different classes. This overlap could make it challenging for models to differentiate between similar-looking classes based on intensity alone.

3. **No Strong Correlation Between Features:**

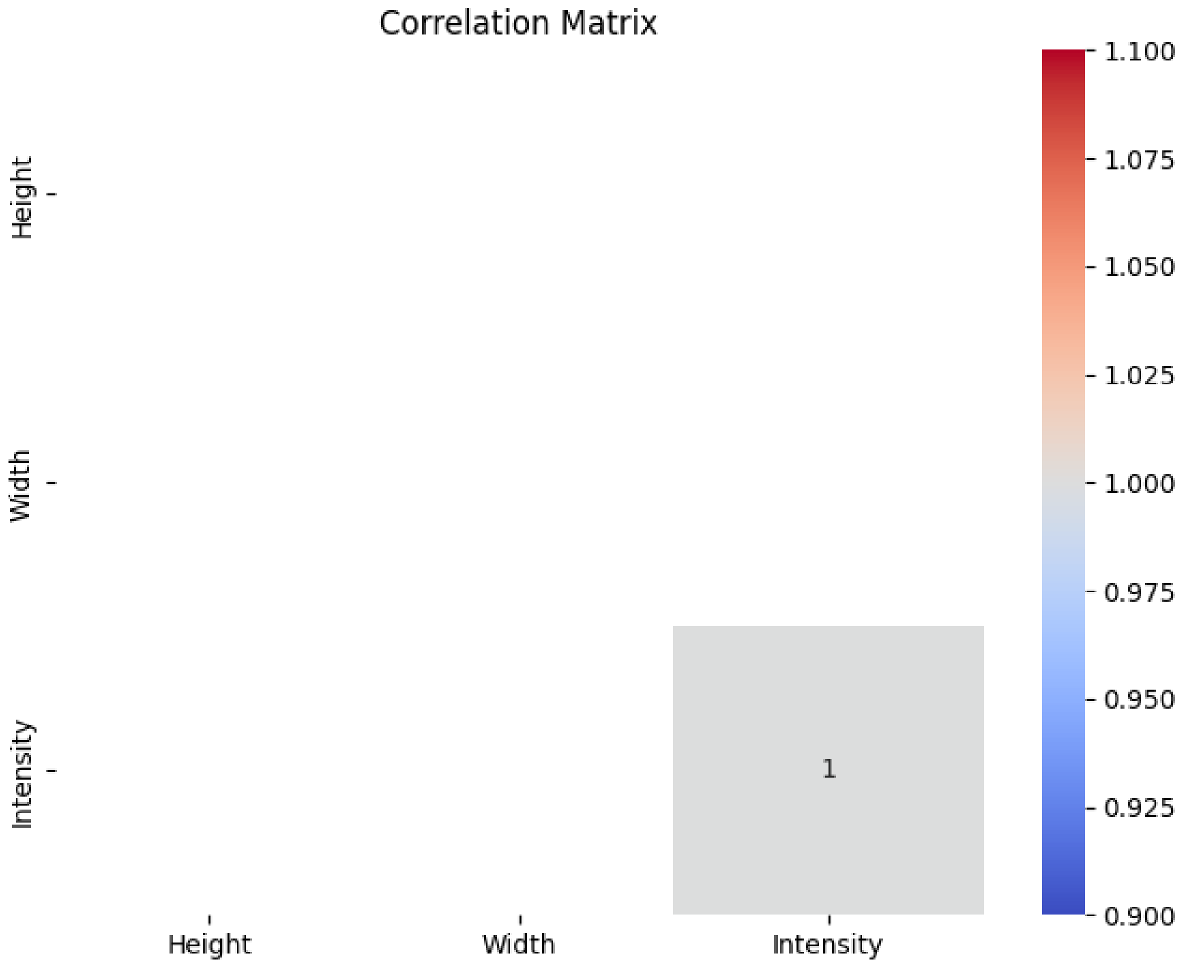
    - The scatter plots show no obvious correlation between Height, Width, and Intensity. This indicates that these features may not be linearly related and suggests the potential need for additional or more complex features to improve class separability.

4. **Height and Width are Constant:**

- Both the Height and Width features show no variability across the dataset. All points are aligned to single values for Height (~520) and Width (~520), indicating that the images have been resized to fixed dimensions during preprocessing.
- Since these features do not vary across the dataset, they are unlikely to contribute to class separability and can be excluded from further analysis.

Correlation Matrix Analysis

The **Correlation Matrix** visualizes the relationships between the features **Height**, **Width**, and **Intensity** in the dataset. It is presented as a heatmap with values indicating the strength of the linear relationships between the features.



Insights and Findings

1. **Constant Height and Width:**
   - The Height and Width features have no visible correlations because they are constant across the dataset, as seen in the pairplot. Since these features do not vary, their correlation with other features cannot be calculated or visualized effectively. This is why the heatmap shows no correlation values for these features.
2. **Self-Correlation of Intensity:**
   - The only visible entry in the heatmap is the self-correlation of Intensity, which is 1.0 (as any feature is perfectly correlated with itself).

- No relationships between Intensity and the other features (Height, Width) are shown because the latter are constant.

3. **Lack of Feature Relationships:**
    - The absence of meaningful correlations reinforces the earlier conclusion that Height and Width do not contribute useful variability to the dataset.

Descriptive Analysis

Descriptive Statistics Report

Objective:
The aim of this analysis is to summarize the key attributes of the images in the dataset, specifically the **Height**, **Width**, and **Average Intensity** of the images.

Methodology:
The dataset consists of 14,377 images. For each image, we extracted three key attributes:

1. **Height**: The vertical size of the image.
2. **Width**: The horizontal size of the image.
3. **Average Intensity**: The mean pixel intensity across the entire image.

The following table presents the descriptive statistics of these attributes:

Attribute	**Height**	**Width**	**Intensity**
Count	14377.0	14377.0	14377.000000
Mean	512.0	512.0	108.365211
Std	0.0	0.0	23.563105

Attribute	Height	Width	Intensity
Min	512.0	512.0	30.144850
25%	512.0	512.0	91.104824
50%	512.0	512.0	106.423073
75%	512.0	512.0	125.031802
Max	512.0	512.0	211.309153

Insights and Findings

- **Height and Width**:

  The height and width of all images in the dataset are consistent, each having a size of **512x512** pixels. The standard deviation for both height and width is **0.0**, which means all images are the same size.

- **Intensity**:

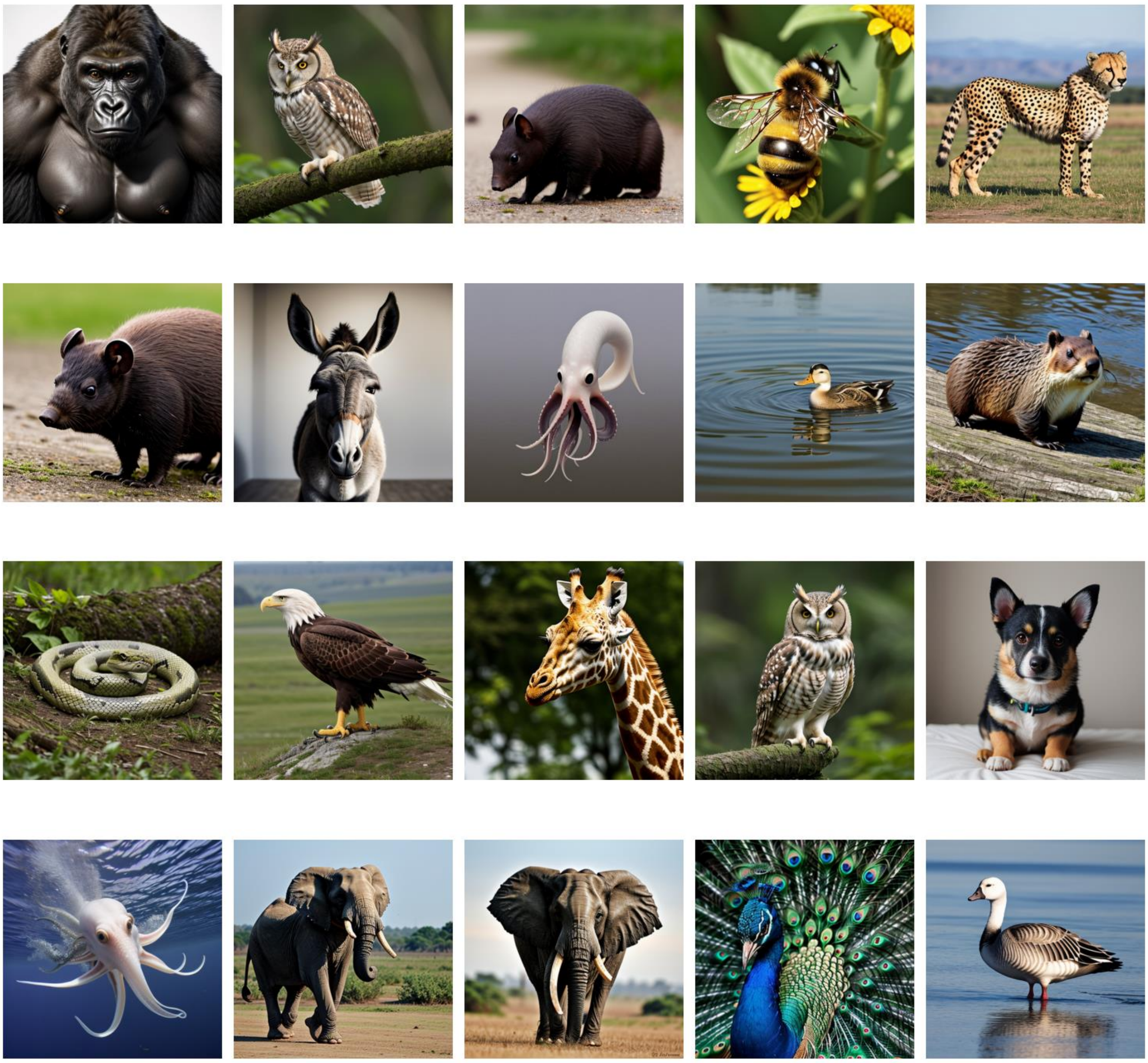
  The **Average Intensity** of the images varies. The mean intensity is **108.37**, with a standard deviation of **23.56**, indicating a moderate level of variability in brightness. The minimum intensity is **30.14**, while the maximum intensity is **211.31**. This range suggests that the images in the dataset vary in terms of brightness, with some being darker and others lighter. The 25th percentile of intensity is **91.10**, the median (50th percentile) is **106.42**, and the 75th percentile is **125.03**, which shows a moderate distribution of intensity values around the mean.

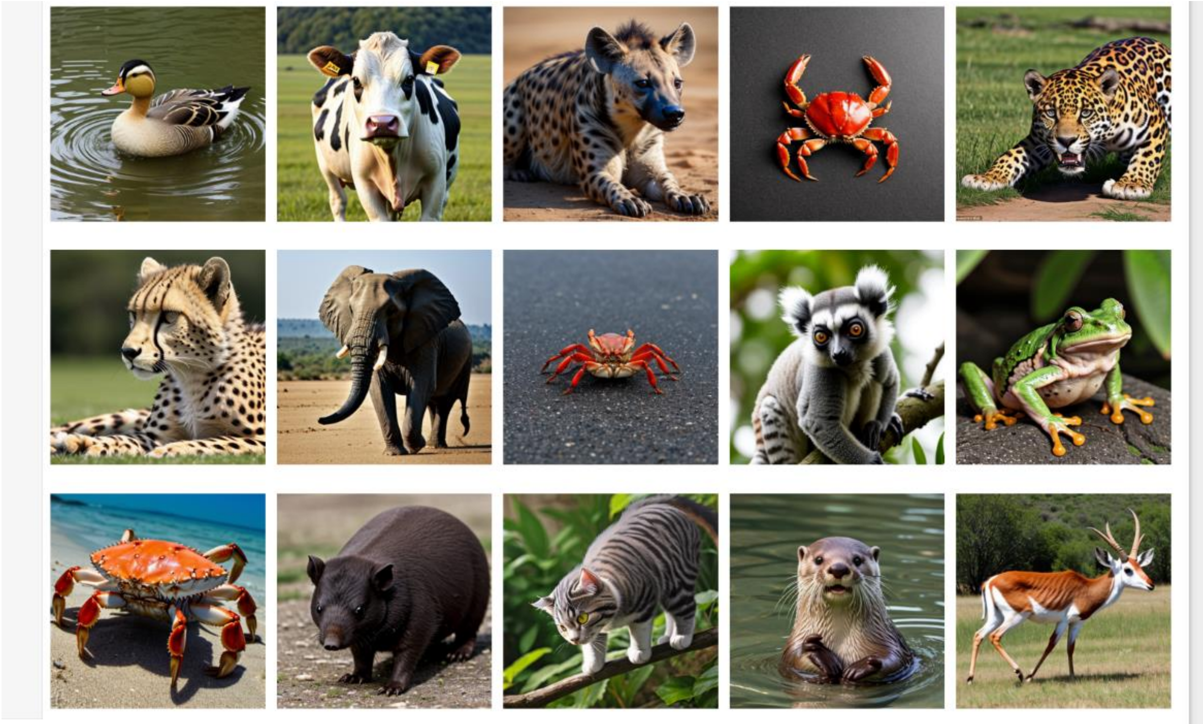
The dataset contains images of uniform size (512x512 pixels). The average intensity values of the images exhibit variability, which may suggest the presence of different lighting conditions or image processing methods used in the dataset. Further analysis could explore the relationship between intensity and other image features, such as colour or texture, to gain more insights into the image properties.

Random Image Visualization by Class

This analysis involves visualizing a random selection of images from the dataset, categorized by their respective classes. The purpose is to ensure the dataset is diverse, well-labeled, and suitable for building robust classification models. Here's why this step is significant:

- **Quality Control:** Visual inspection can reveal labelling errors, missing data, or corrupted images.
- **Dataset Diversity:** Identifying class imbalances or homogeneity in the data is crucial for model performance.
- **Feature Insights:** Observing patterns or features in images can guide feature extraction and model design.





Insights and Findings

1. **High Diversity in Certain Classes:**
   Some classes exhibit a wide range of visual variability, including differences in background, pose, and lighting. This variability can enrich the model's learning process but might require more data to ensure generalization.
2. **Consistent Features in Some Classes:**
   Some classes display relatively uniform features like consistent colour tones or predictable patterns. This consistency can aid classification but might result in overfitting if the model relies solely on these features.
